Luke Rodel

Mechanical design · Automation · Manufacturing

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Hands-on engineering that connects mechanical design to working systems.

Mechanical engineering student at the University of Alabama. Interested in robotics, testing, and automation.
Expected graduation: May 2027.

01 / SPUDNIK EQUIPMENT · SUMMER 2024

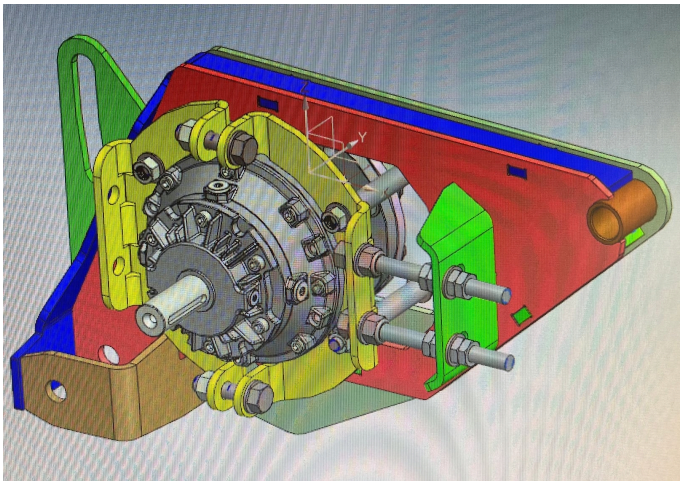
Mechanical design & reliability

A simpler mount. A more reliable drivetrain.

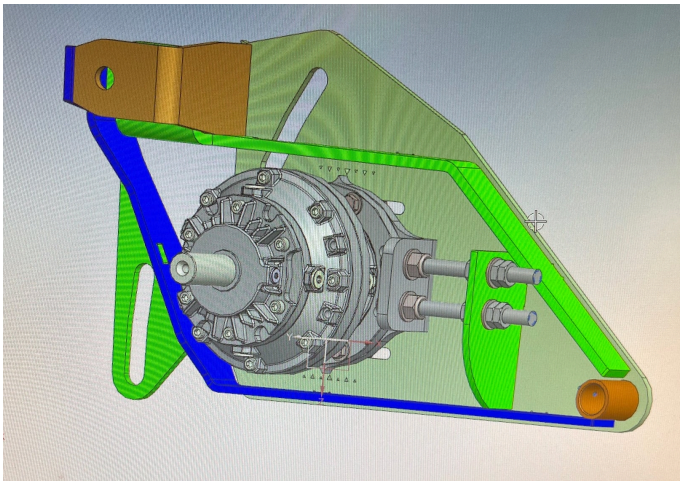
I redesigned a sugar beet harvester's chain-drive layout and mounting arrangement to improve tensioning, assembly, and service access.

10 fewer parts

in the redesigned assembly



BEFORE Original assembly



AFTER My simplified design

The decisions

I investigated chain skipping and whip, iterated layouts in NX, raised the chain's low point, and incorporated tensioning arms. Simplifying the mounting arrangement also addressed assembly access around the gearbox.

The validation

After implementation, the system ran at speed without the previously observed chain skipping. The website includes the original running clip and more detail about my role and testing observations.

NX CAD · Design for assembly · Chain tensioning · Hardware validation

University of Alabama

B.S. Mechanical Engineering candidate
Robotics minor candidate · GPA 3.96 / 4.0

Currently building

Senior capstone: robotic-arm direct ink writing system for non-planar polymer lattices. In progress.

Luke Rodel

Selected engineering work

Automation & production systems

02 / MERCEDES-BENZ · FALL 2024

Machine vision & integration



Full angled printer installation.

Inspection without the extra steps.

I tested scanning angles and designed an angled printer stand to reduce glare. Working alongside the robotics lead, I helped integrate automatic grading as labels printed.

≈10 seconds

saved in the label workflow

PRINT → INSPECT → ACCEPT OR REPRINT

03 / MERCEDES-BENZ · SUMMER 2025

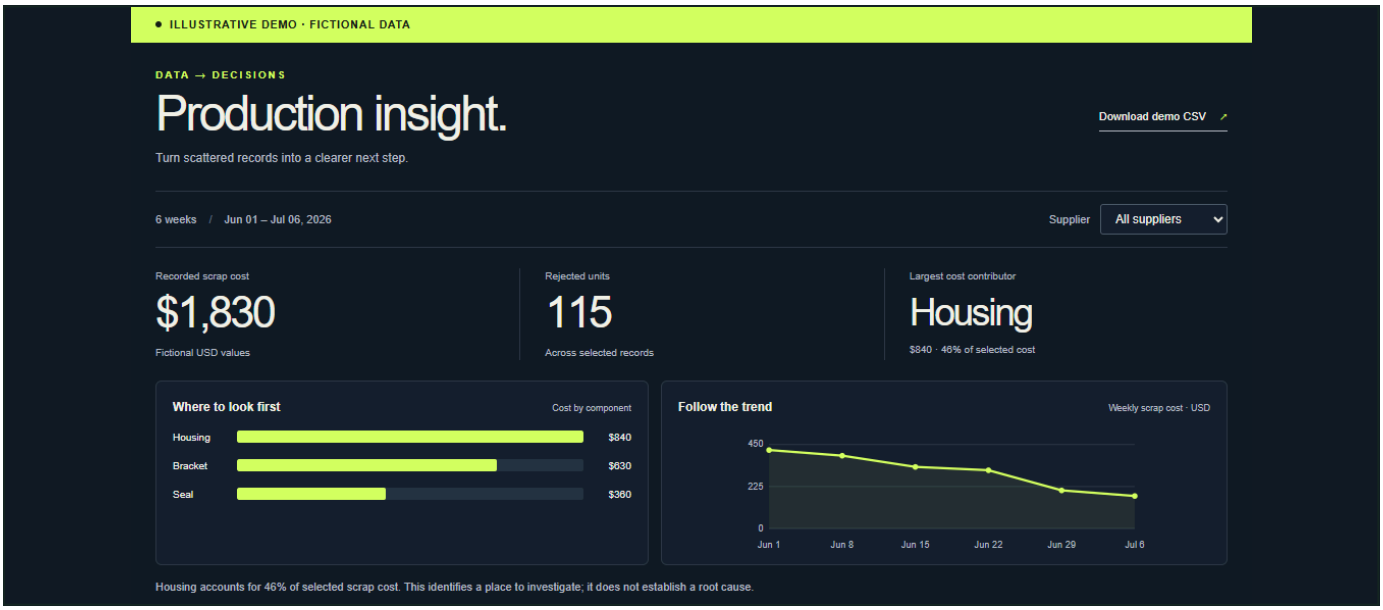
Data analysis & workflow automation

A clearer picture of production.

I built Power BI reporting tools and mapped defect, repair, reorder, and chargeback workflows across departments.

Hours → minutes

for a separate monthly reporting workflow



Independent web recreation with fictional data. This is not the employer's Power BI report.

I used Power Query / M and DAX to prepare data and improve visibility. A related process investigation informed a proposal for better scrap and reclaim traceability; it is distinct from an implemented plant-wide system or a verified financial saving.

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Packaging, testing & hands-on engineering

Selected work · 2024–2026

HERRICK TECHNOLOGY LABORATORIES · SUMMER 2026

Mechanical Engineering Intern

Radio hardware. Real-world conditions.

At a defense radio contractor, I worked on mechanical packaging and testing for direction-finding antennas and radio card assemblies.



Direction-finding antenna family



3U VPX radio card family

Public product imagery from Herrick Technology Laboratories: DF antenna family · 3U VPX family. My contribution was to mechanical packaging and testing.

Package & protect

I updated drawings, CAD models, and BOMs; coordinated Altium board layouts with mechanical packaging; prepared hardware for drop and immersion tests; and designed custom carrying cases with calculated foam inserts.

Test & refine

I performed physical thermal tests, calculations, and Autodesk CFD simulations to compare parts and support thermal testing standards. I also evaluated size, weight, power, and cost, and automated insert details in Creo.

MORE FROM MERCEDES-BENZ · 2024–2025

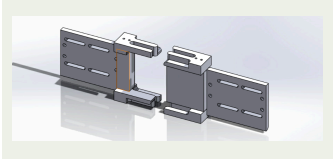
Maintenance & production systems

Mechanical design on the line

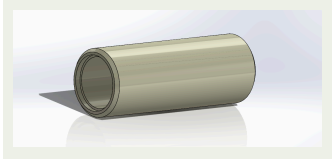
I designed and fabricated front-end support brackets, reverse engineered replacement parts, created roller nestings and FIFO storage, and designed scanner mounts, tablet hand rests, and a mobile camera cart.

Workflows & practical tools

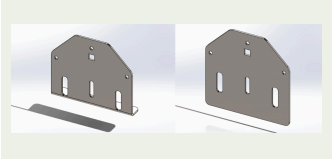
Additional work included label jigs and work instructions, error-log comparison tools, HV-cap maintenance, production layout updates, reporting automation, and AI tools and training with the Data Team.



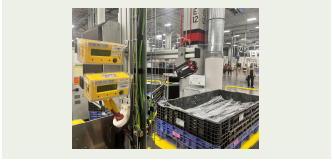
Support brackets



Replacement roller



Mounting plates



Scanner integration

Explore the full project notes, running drivetrain video, interactive production demo, and parts gallery at lukerodel.github.io/luke-rodel-portfolio.